

7(a)	any four from: 1 insulin / it, is a <u>hormone</u> ; 2 (described) stimulus causes insulin to be secreted from (named) endocrine, gland / tissue / cells ; 3 insulin travels in blood ; 4 binds to receptors on (named) target <u>cells</u> ; 5 to cause (described) response ; 6 AVP ;	4
7(b)(i)	any four from: 1 at 21 min, blood glucose concentration / it, rises to $110 \text{ mg } 100 \text{ cm}^{-3}$; 2 (high / increase in, blood glucose concentration) detected by pancreas / causes response in pancreas ; 3 insulin rises to 500 pmol dm^{-3} ; 4 (insulin) increase GLUT proteins in membrane / increases glucose uptake into cells / increases cell permeability to glucose or (insulin) increases rate of respiration of glucose ; 5 (insulin) causes / stimulates, glycogenesis / lipogenesis ; 6 (mean) blood glucose concentration / it, falls, to normal / within normal range / below starting point / to set point ;	4

Question	Answer	Marks
7(b)(ii)	any three from: 1 blood glucose concentration falls as glucose is respired ; 2 blood glucose concentration falls below, set point / normal ; 3 insulin, secretion / release, stops ; 4 <u>glucagon</u> (secreted / released / rises) ; 5 glycogen, (in cells) converted to glucose or glycogenolysis ; 6 (stored) protein / fat, converted to glucose or gluconeogenesis or (stored) protein / fat, used for respiration ;	3

Question	Answer	Marks
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